Scientific computing II 2026

Exercise 2
Submit your solution to Moodle no later than Thursday 12.11.2026 23:59
All Fortran programs mentioned below are given in Moodle (in either this or previous exercises). For all problems, maximum is 2 points.

*Note: The practical way to define the integer and real **kinds** is to use the* `iso_fortran_env` *module. See the example in the attached file* `iso_fortran_env_kinds.f90` *in the package* `ex02_codes.zip`*.*

Problem 1.

Write a program that reads in an integer `n` from user and writes to screen the squares of integers from 2 to `n`.

Problem 2.

What are the iteration counts of the following do loops, the values of loop iteration variable `i` inside the loop, and its value after the do construct? Write a program that does it for you.

```
do i=0,5
do i=5,1,-1
do i=10,1,-2
do i=0,30,7
```

Problem 3. (1 point for (a) and (b) each)

(a) Compile and run the program `subtract.f90` as it is given. Then comment out on lines 4 and 5 the kind specifiers (`_rk`). Explain the difference in the outputs of these two cases.
(b) Compile and run the program `expression_equivalence.f90`. Program calculates expression $x^2 - y^2$ using three different expressions that are *mathematically* equivalent:

$(x - y)(x + y)$
$x^2 - y^2$
$xx - yy$

Explain why the output is what it is.

Problem 4. (1 point for (a) and (b) each)

(a) Modify the attached program `testintkind.f90` and check what real kinds your compiler/hardware supports. The function used to select the kind wants two arguments: `selected_real_kind(p,r)`. You can set the last argument (range as decimal exponent, `r`) to something reasonable (like 20) and scan over the values of precision `p`.
(b) Modify the program `integerkind.f90` (shown on lecture notes 'Integer and

real kinds', slide 7) so that instead of 2 it multiplies variable `i` by 3 in every iteration. Explain the difference with the original program and reason for it.

Both program are also included in the attached package `ex02_codes.zip`.

Problem 5. (1 point for (a) and (b) each)

(a) Using the results of problem 4(a), find out using the floating point inquiry functions (lecture notes 'Integer and real kinds', slide 16) what are the properties *largest number* and *smallest positive number* of these floating point kinds. The functions to be used all take one argument as a floating point variable of the desired kind. i.e. if you want to find out the largest number that can be presented by variable `x` (that has a certain kind), use the function call `huge(x)`.
(b) Machine epsilon is defined as the smallest positive number that when added to one produces a number that is greater than one. There is a function for this: `epsilon(x)` that returns the machine epsilon for the same kind of floating point number as `x` is. Demonstrate that this really is so: write a short program that first adds `epsilon(x)` to 1.0 and `epsilon(x)/2.0` to 1.0. What do you get if you instead of addition subtract this number from 1.0? Be careful to do all assignments and arithmetics with the desired kinds.

Problem 6.

Calculate the harmonic sum

$$H_m = \sum_{n=1}^{m} \frac{1}{n}$$

using all the floating point kinds you found in problem 4. Do the summation up to $m = 10^7$. Comment on the differences you found in the results.

Note1: In order to get the inverse of integer `n` *inside the loop evaluated as a floating point number with kind* `rk`*, do it like* `1.0_rk/real(n,rk)`*, where* `rk` *is the kind of the floating point variable you are using.*
Note2: In order to make the execution faster use option `-Ofast` *(or* `-fast` *in case you use Intel compiler) when compiling your code..*